Akoizo Original Practice - Form 2 - Answer key

Division B | Botany | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which tissue primarily carries water and minerals?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

A. Epidermal hairs only

B. Pollen

C. Xylem

D. Phloem only

Correct: C

2. (2 points) A plant spore is produced by…

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

A. Mitosis of a diploid egg only

B. Pollination alone

C. Meiosis in the sporophyte

D. Fertilization of two seeds

Correct: C

3. (2 points) Rotating branches at a node changes relationships?

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

A. Only for plants

B. Only for fossils

C. No

D. Yes

Correct: C

4. (2 points) Yellow leaves uniquely prove nitrogen deficiency?

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

A. Yes

B. Only in all monocots

C. Only at noon

D. No

Correct: D

5. (2 points) Tall-tree xylem flow is generally explained by…

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

A. Each vessel pumping with ATP

B. Phloem always pushing upward

C. Gravity pulling water upward

D. Cohesion-tension linked to transpiration

Correct: D

6. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Photosynthesis and plant energy.
Worked-case data: Low light: 5 bubbles per minute. Medium light: 10 bubbles per minute. High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.
Separate investigation observations:
Low light: 5 bubbles per minute.
Medium light: 10 bubbles per minute.
High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.

answer: Compare low and medium light: the indicator increases. Compare medium and high light: the indicator stays constant. Test another limiting factor while keeping the method consistent. (4 points).

7. (2 points) Natural origin guarantees medicinal safety?

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

A. Yes

B. Only for leaves

C. Only for roots

D. No

Correct: D

8. (2 points) A thicker cuticle commonly helps reduce…

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

A. Uncontrolled water loss

B. All respiration

C. Every nutrient deficiency

D. All need for roots

Correct: A

9. (2 points) “Cryptogams” is best treated as…

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

A. A historical descriptive grouping

B. One exclusive modern clade

C. A synonym for all angiosperms

D. A seed tissue

Correct: A

10. (2 points) Changing water and nutrients together reveals the unique cause?

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

A. No

B. Yes

C. Only if leaves become greener

D. Only at high light

Correct: A

11. (4 points) Distinguish wall and membrane.

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

answer: The wall provides structural support outside the selectively permeable plasma membrane. (4 points).

12. (4 points) Track n and 2n in the example.

answer: 2n sporophyte -> meiosis -> n spore -> n gametophyte -> n gamete -> fertilization -> 2n zygote. (4 points).

13. (4 points) Why is a tree not a progress ladder?

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

answer: It shows branching ancestry among lineages, not a ranking of living organisms by superiority. (4 points).

14. (4 points) Why do plants need respiration?

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

answer: Cells use chemical energy to produce ATP and support metabolism; photosynthesis does not eliminate cellular respiration. (4 points).

15. (4 points) Why distinguish phloem and xylem mechanisms?

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

answer: They transport different materials and use different gradients and cellular structures. (4 points).

16. (4 points) Name a terrestrial tradeoff.

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

answer: A cuticle reduces water loss but restricts direct exchange, favoring regulated gas pathways. (4 points).

17. (4 points) Name two plant ecosystem functions.

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

answer: Examples include primary production, habitat structure, erosion reduction, water exchange, and nutrient cycling. (4 points).

18. (4 points) What does a directional-light control test?

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

answer: Whether curvature differs under changed light geometry while other conditions are comparable. (4 points).